

Semantic-Aware Integrated Sensing and Communication (ISAC) Networks with Power-Domain NOMA and 3D Elevation-Dependent Path Loss

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Abstract

Integrated Sensing and Communication (ISAC) has emerged as a key technology for 6G wireless systems to share spectrum and hardware resources efficiently. However, resource scarcity and security vulnerability in Air-to-Ground (A2G) networks pose critical implementation challenges. In this report, we present the design and system model of a Semantic-Aware ISAC Network assisted by an aerial Reconfigurable Intelligent Surface (RIS) mounted on an Unmanned Aerial Vehicle (UAV) (R). To optimize spectral efficiency, we employ Power-Domain Non-Orthogonal Multiple Access (PD-NOMA) to serve a near user (Target T , which is the updated form of the vehicle node) and a distant mobile user (U) simultaneously. Crucially, all communications and sensing nodes operate under semantic-aware metrics, shifting the focus from error-free bit transmission to the preservation of meaning. To capture realistic aerial propagation, we integrate a 3D elevation-dependent path loss model that dynamically adjusts the Line-of-Sight (LoS) probability and Rician K -factor. Additionally, the system is subjected to directional jamming from a UAV Jammer (J), which also performs monostatic radar sensing. Eavesdropping nodes (E_1, E_2, E_3) are distributed to wiretap the downlink transmissions. We formulate the detailed mathematical models and signal flows for the entire network, providing a rigorous foundation for subsequent optimization and reinforcement learning studies.

Index Terms

Integrated Sensing and Communication (ISAC), Semantic Communications, Power-Domain NOMA, Reconfigurable Intelligent Surface (RIS), Air-to-Ground (A2G) Path Loss, Successive Interference Cancellation (SIC).

I. INTRODUCTION

Future 6G networks are envisioned to go beyond traditional communication by integrating radar sensing and semantic intelligence into a unified architecture. Integrated Sensing and Communication (ISAC) allows wireless nodes to simultaneously transmit information and sense environment parameters (e.g., target localization and tracking) using the same frequency bands and hardware.

To overcome severe blockages and path loss in urban environments, Reconfigurable Intelligent Surfaces (RIS) mounted on Unmanned Aerial Vehicles (UAVs) (R) can be deployed as passive aerial reflectors. By adjusting the phase shifts of the RIS elements, a virtual Line-of-Sight (LoS) link can be established between a terrestrial Base Station (BS) B and ground users, dramatically increasing signal coverage and physical layer security.

In parallel, semantic communications represent a paradigm shift from Shannon's classical information theory, which focuses solely on the accurate transmission of bit sequences. Rather than guaranteeing bit-level fidelity, a semantic-aware transmitter extracts the task-relevant meaning of a message through a learned semantic encoder and conveys only a compact set of semantic symbols that are sufficient for the intended receiver to reconstruct that meaning. This approach drastically reduces the required channel symbols and increases resilience against severe fading and interference, while simultaneously offering an inherent security advantage: an intercepted semantic symbol stream is largely uninterpretable to any node that does not possess the same semantic encoder-decoder pair, since meaning recovery – unlike bit recovery – depends on shared semantic knowledge rather than on the raw symbols themselves.

This report outlines the implementation details and mathematical models completed during Week 9. We design a semantic-aware ISAC network incorporating:

- 1) **Elevation-Dependent Path Loss:** Incorporating 3D elevation angle θ_{ij} , sigmoidal LoS probability $P_{\text{LoS}}(\theta_{ij})$, and elevation-dependent Rician fading.
- 2) **Power-Domain NOMA (PD-NOMA):** Serving a near Target user (T) and a distant Mobile User (U) over a superposed downlink signal, using semantic-aware Successive Interference Cancellation (SIC), in which the near user first reconstructs the far user's semantic information and removes its contribution before decoding its own semantic symbols.

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- 3) **Semantic Performance Metrics:** Characterizing semantic similarity, semantic rate, and semantic distortion for downlink, uplink, V2X, and eavesdropper channels – each governed by the semantic compression ratio between semantic concepts and transmitted channel symbols – as well as semantic sensing accuracy at the radar receiver.

No simulation plots are provided in this weekly report, as the focus is solely on the mathematical model definition and software implementation architecture.

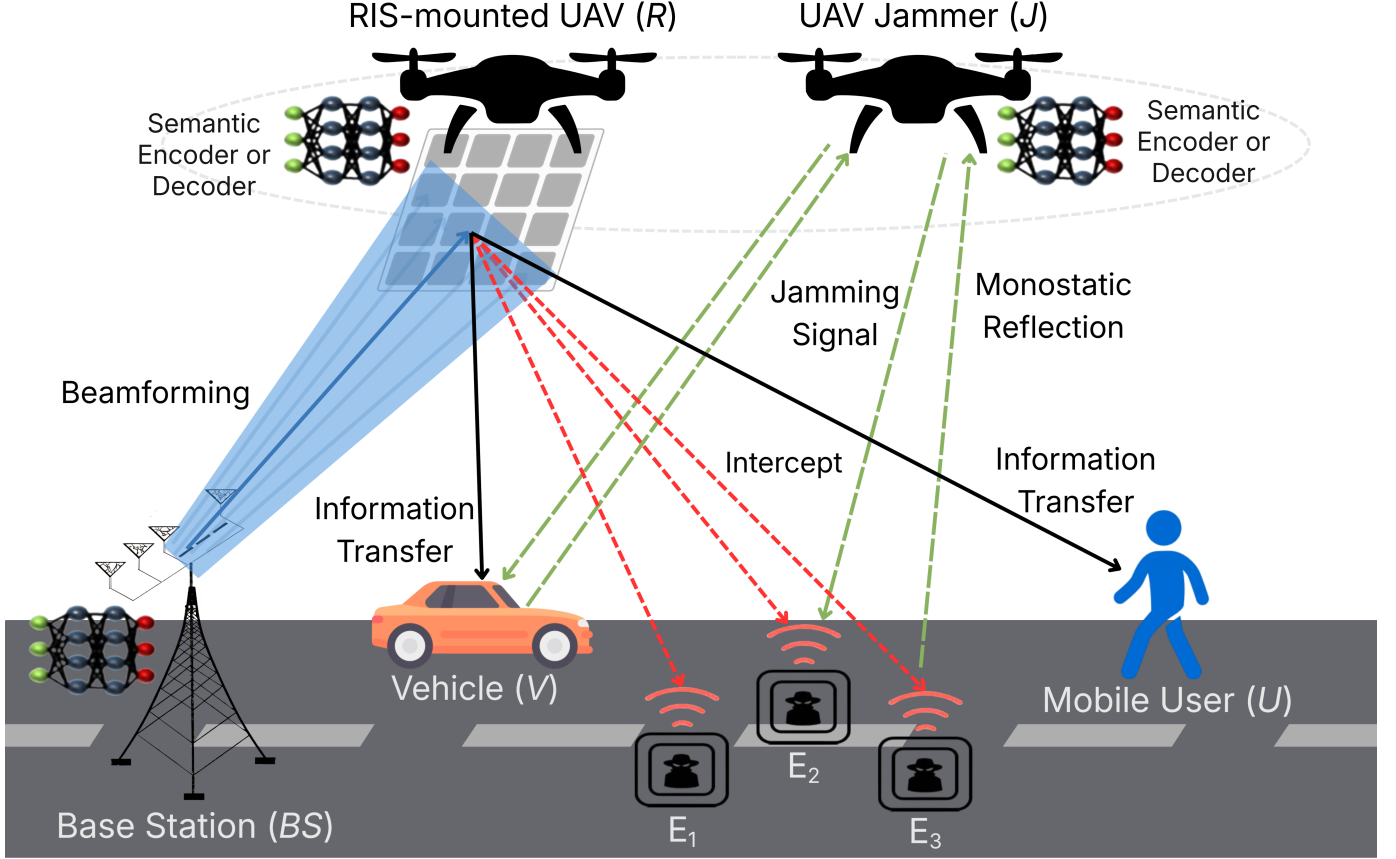


Fig. 1. System model of the proposed semantic-aware RIS-assisted UAV ISAC network. The Base Station (BS) communicates with the Target (T) and Mobile User (U) through an RIS-mounted UAV (R) using beamforming and PD-NOMA. A UAV Jammer (J) transmits jamming signals while performing monostatic radar sensing reflections off the RIS R . Semantic encoders and decoders are employed at the communicating nodes.

II. SYSTEM ARCHITECTURE AND NODE DESCRIPTIONS

We consider an aerial-terrestrial co-existing network operating in a bounded 3D space. The system consists of the following key nodes:

- **Base Station (B):** Located at $\mathbf{p}_B = [x_B, y_B, z_B]^T$. It acts as a semantic-aware transmitter sending superposed downlink signals to paired users.
- **RIS-mounted UAV (R):** An aerial platform hovering at $\mathbf{p}_R = [x_R, y_R, z_R]^T$, equipped with N_R passive reflecting elements to establish cascaded communication paths.
- **UAV Jammer (J):** An adversarial aerial node located at $\mathbf{p}_J = [x_J, y_J, z_J]^T$. It transmits directional jamming signals to degrade ground user rates while performing monostatic radar sensing reflections off the RIS R .
- **Target (T):** A near user and active communicator (representing the updated form of the vehicle) located at $\mathbf{p}_T = [x_T, y_T, z_T]^T$. In the downlink, it decodes the NOMA signal using semantic-aware SIC. In the uplink, it transmits semantic messages to the Base Station B (via the RIS R) and directly to the Mobile User U via V2X.
- **Mobile User (U):** A distant user located at $\mathbf{p}_U = [x_U, y_U, z_U]^T$, decoding the downlink far-user signal directly.
- **Eavesdroppers (E_1, E_2, E_3):** Ground nodes located at positions $\mathbf{p}_{E_i}$, trying to intercept the downlink semantic transmissions.

III. MATHEMATICAL FORMULATIONS

Before detailing the individual link models, we first outline the end-to-end semantic communication pipeline that underlies every transmission in the network, from the Base Station downlink to the Target's uplink and V2X links. A semantic message

originating at a *semantic source* (e.g., a sentence, a sensor reading, or a control instruction) is first mapped by a *semantic encoder*, implemented as a joint source-channel coding (JSCC) neural network, into a sequence of M semantic concepts. These concepts are subsequently represented by K transmitted channel symbols, where K is generally much smaller than the number of bits required for a conventional, meaning-agnostic bit-level representation of the same message. The resulting semantic symbol stream is transmitted over the elevation-dependent wireless channel described in Section III-A, which introduces path loss, small-scale fading, jamming interference, and, where applicable, eavesdropping. At the legitimate receiver, a matched *semantic decoder* maps the received (and possibly noisy) symbols back into an estimate of the original meaning. Formally, this pipeline can be summarized as

Semantic Source $\rightarrow$ Semantic Encoder (JSCC) $\rightarrow$ Semantic Symbols $\rightarrow$ Wireless Channel $\rightarrow$ Semantic Decoder $\rightarrow$ Recovered Meaning

The ratio M/K , referred to as the *semantic compression ratio*, governs the trade-off between transmission efficiency and semantic fidelity: a larger M/K conveys more semantic concepts per transmitted symbol (higher efficiency) but places greater reconstruction burden on the semantic decoder under a given channel quality, generally increasing susceptibility to semantic distortion; a smaller M/K trades efficiency for robustness. This compression ratio directly scales the semantic rate expression introduced in Section III-C and is treated identically across all downlink, uplink, V2X, and eavesdropper links considered in this report.

To make this efficiency-reliability trade-off explicit at the physical layer rather than leaving it as a purely descriptive statement, every received SINR that feeds a semantic decoder in this report (Sections III-B, III-D, and III-F) is defined as a *semantic-effective SINR*: the conventional link-budget SINR (transmit power scaled by channel gain, divided by jamming interference plus noise) is additionally scaled by the inverse compression ratio K/M . Formally, for any link with conventional link-budget ratio Γ (i.e., the power-domain signal-to-interference-plus-noise ratio computed from transmit power, channel gain, jamming power, and σ^2 in the usual manner),

$$\text{SINR} = \left(\frac{K}{M} \right) \cdot \Gamma, \quad (1)$$

where $K/M \leq 1$ under net compression ($M \geq K$). The rationale is that when M semantic concepts share the protection of only K transmitted channel symbols, the effective reliability apportioned to each individual concept is diluted in proportion to the number of symbols available per concept: aggressive compression (M/K large) reduces the semantic-effective SINR relative to the raw link-budget ratio Γ , even though the physical channel and transmit power are unchanged, whereas conservative compression ($M/K \rightarrow 1$) recovers the conventional SINR. This definition is applied consistently to every SINR expression in the remainder of the report; the underlying link-budget ratio Γ for each specific link is given explicitly in the corresponding subsection.

A. Elevation-Dependent Air-to-Ground Path Loss and Channel Model

For any propagation link ij between a transmitter i at position $\mathbf{p}_i = [x_i, y_i, z_i]^T$ and a receiver j at $\mathbf{p}_j = [x_j, y_j, z_j]^T$ (with $ij \in \{BR, RT\}$ denoting the links from BS B to RIS R and RIS R to Target T , respectively), the 3D Euclidean distance is:

$$d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2 = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}. \quad (2)$$

The 2D horizontal distance is $d_{2D,ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$. The vertical elevation difference is $\Delta z_{ij} = |z_i - z_j|$. The 3D elevation angle θ_{ij} (in degrees) is given by:

$$\theta_{ij} = \arcsin \left(\frac{\Delta z_{ij}}{d_{ij}} \right) \cdot \frac{180}{\pi} = \arctan \left(\frac{\Delta z_{ij}}{d_{2D,ij}} \right) \cdot \frac{180}{\pi}. \quad (3)$$

Due to the variations in building heights, blockages, and terrain, the probability of establishing a Line-of-Sight (LoS) link is modeled as a sigmoidal function of the elevation angle θ_{ij} :

$$P_{\text{LoS}}(\theta_{ij}) = \frac{1}{1 + a \cdot \exp(-b(\theta_{ij} - a))}, \quad (4)$$

where a and b are calibration constants representing the urban environment (e.g., $a = 9.61, b = 0.16$ for typical urban environments). The Non-Line-of-Sight (NLoS) probability is $P_{\text{NLoS}}(\theta_{ij}) = 1 - P_{\text{LoS}}(\theta_{ij})$.

The path losses for the LoS and NLoS states, incorporating excessive attenuation, are modeled as:

$$\text{PL}_{\text{LoS}}(d_{ij}) = 20 \log_{10} \left(\frac{4\pi d_{ij} f_c}{c} \right) + \eta_{\text{LoS}}, \quad (5)$$

$$\text{PL}_{\text{NLoS}}(d_{ij}) = 20 \log_{10} \left(\frac{4\pi d_{ij} f_c}{c} \right) + \eta_{\text{NLoS}}, \quad (6)$$

where f_c is the carrier frequency, c is the speed of light, and η_{LoS} , η_{NLoS} represent the excessive path loss factors in dB for LoS and NLoS links, respectively. The mean path loss in dB is:

$$\overline{\text{PL}}(d_{ij}, \theta_{ij}) = P_{\text{LoS}}(\theta_{ij}) \cdot \text{PL}_{\text{LoS}}(d_{ij}) + (1 - P_{\text{LoS}}(\theta_{ij})) \cdot \text{PL}_{\text{NLoS}}(d_{ij}). \quad (7)$$

Converting this to linear gain yields:

$$\text{PL}_{\text{linear}}(d_{ij}, \theta_{ij}) = 10^{-\overline{\text{PL}}(d_{ij}, \theta_{ij})/10}. \quad (8)$$

The small-scale fading envelope power gain is represented by h throughout the paper (replacing f). For Rayleigh fading, the fading power gain is exponentially distributed with unit mean, i.e., $h \sim \text{Exp}(1)$. For Rician fading, we model the fading power gain as:

$$h = |g_{\text{fade}}|^2, \quad g_{\text{fade}} = \sqrt{\frac{K}{K+1}} \bar{g}_{\text{fade}} + \sqrt{\frac{1}{K+1}} \tilde{g}_{\text{fade}}, \quad (9)$$

where $\bar{g}_{\text{fade}}$ is the deterministic LoS component, $\tilde{g}_{\text{fade}} \sim \mathcal{CN}(0, 1)$ is the scattered NLoS component, and K is the Rician K -factor. In our model, K scales with the elevation angle:

$$K_{\text{dB}}(\theta_{ij}) = K_0 + \alpha_K \theta_{ij}, \quad K(\theta_{ij}) = 10^{K_{\text{dB}}(\theta_{ij})/10}, \quad (10)$$

where $K_0 = 3.0$ dB and $\alpha_K = 0.15$ dB/degree. The final channel power gain g_{ij} is represented as:

$$g_{ij} = \text{PL}_{\text{linear}}(d_{ij}, \theta_{ij}) \cdot h_{ij}, \quad (11)$$

where h_{ij} represents the small-scale fading gain of link ij .

B. Power-Domain NOMA and Semantic-Aware Successive Interference Cancellation

In the downlink phase, the Base Station B transmits a superposed signal to serve both the near user (Target T) and the far user (Mobile User U) using the same time-frequency resource. Under semantic communications, the signals s_N and s_F represent the semantic symbol sequences generated by deep learning joint source-channel coding (JSCC) encoders for the Target T and Mobile User U , respectively – that is, each of s_N and s_F already embodies the compressed semantic representation of its respective source message, following the pipeline introduced at the start of Section III, rather than a conventional bit-level codeword:

$$s = \sqrt{a_N P_B} s_N + \sqrt{a_F P_B} s_F, \quad (12)$$

where P_B is the Base Station transmit power, and a_N, a_F are the power allocation factors satisfying:

$$a_N + a_F = 1, \quad a_F > a_N \geq 0. \quad (13)$$

The signals propagate through two primary links: the direct link from the BS B and the reflected cascaded link via the RIS R . The cascaded channel gain between the Base Station at position $\mathbf{p}_B$ and a destination user $k \in \{T, U\}$ at position $\mathbf{p}_k$ through the RIS R at $\mathbf{p}_R$ is computed as:

$$g_{\text{cascaded}, k} = g_{B,R} \cdot g_{R,k} \cdot N_R^2, \quad k \in \{T, U\}, \quad (14)$$

where $g_{B,R} = \text{PL}_{\text{linear}}(d_{B,R}, \theta_{B,R}) \cdot h_B$ (with h_B representing small-scale fading on the BS-RIS link) and $g_{R,k} = \text{PL}_{\text{linear}}(d_{R,k}, \theta_{R,k}) \cdot h_R$ (with h_R representing small-scale fading on the RIS-destination link) are the individual elevation-dependent channel gains, and N_R is the number of RIS reflecting elements. The total effective channel gain for user $k \in \{T, U\}$ is:

$$g_k = g_{\text{cascaded}, k} + g_{\text{direct}, k}. \quad (15)$$

For decoding, the users are ordered based on their effective channel gains. If $g_T < g_U$ due to shadowing or user mobility, the roles are swapped dynamically. Under normal conditions:

$$g_T \geq g_U. \quad (16)$$

The UAV Jammer J emits artificial noise with power P_J , resulting in received jamming interference power at receiver $k \in \{T, U\}$:

$$P_{J,k} = P_J \cdot g_{J,k}, \quad (17)$$

where $g_{J,k}$ is the elevation-dependent channel gain from the Jammer J to receiver k . Note that for the Target T , the jamming interference is specifically denoted as $P_{J,T}$ (replacing the symbol $I_{\text{Jam}, \text{Veh}}$).

1) *Distant (Far) User Semantic Decoding*: The Far User (Mobile User U) applies its semantic decoder directly to the superposed received signal to recover s_F , treating the near user's semantic symbols s_N as interference rather than attempting to separate them. Because U operates at a lower effective channel gain, it is allocated the larger power fraction a_F so that its semantic symbols remain decodable at the required fidelity. The conventional link-budget ratio at the Far User is $\Gamma_F = \frac{a_F P_{BGU}}{a_N P_{BGU} + P_{J,U} + \sigma^2}$, where $P_{J,U} = P_J g_{J,U}$ is the jamming power received at Mobile User U , and σ^2 is the background thermal noise power. Applying the semantic-effective scaling introduced above, the semantic SINR at the Far User is:

$$\text{SINR}_F = \left(\frac{K}{M} \right) \cdot \frac{a_F P_{BGU}}{a_N P_{BGU} + P_{J,U} + \sigma^2}, \quad (18)$$

so that a more aggressive compression ratio (M/K large) directly lowers SINR_F even under otherwise identical power, channel, and jamming conditions. This semantic SINR is passed directly through the semantic decoder's operating characteristic: the Mobile User evaluates the recovered meaning using its semantic similarity score $S_U(\text{SINR}_F)$ and semantic rate $R_{\text{sem},U}$ as detailed in Section III-C, rather than a conventional bit-error-rate criterion.

2) *Near User Semantic-Aware SIC Decoding*: The Near User (Target T) employs Successive Interference Cancellation (SIC) at the level of *semantic representations* rather than raw bit streams. Concretely, T first uses its own copy of the far-user semantic decoder to reconstruct an estimate of the far user's semantic information $\hat{s}_F$ from the superposed signal; this reconstructed semantic component is then subtracted from the received signal before T applies its own semantic decoder to recover s_N . Because the far user's semantic contribution is removed prior to decoding, s_F (or its residual) does not act as interference once cancellation is complete. The conventional link-budget ratio at the Target after cancellation is $\Gamma_N = \frac{a_N P_{BGT}}{P_{J,T} + \sigma^2}$, where the denominator contains only the received Jammer interference $P_{J,T}$ and the background noise variance σ^2 . Applying the same semantic-effective scaling, the received SINR at the Target T for decoding its own signal s_N simplifies to:

$$\text{SINR}_N = \left(\frac{K}{M} \right) \cdot \frac{a_N P_{BGT}}{P_{J,T} + \sigma^2}. \quad (19)$$

Note that, since cancellation is performed on the semantic representation rather than on individual bits, any residual imperfection in reconstructing $\hat{s}_F$ manifests as semantic distortion rather than a conventional bit error, and is implicitly captured through the semantic similarity function of Section III-C when evaluated at SINR_F ; the K/M scaling in SINR_N additionally ensures that the same compression ratio governing the far-user reconstruction also degrades the effective reliability of the near user's own SIC-decoded symbol, since both users' semantic streams are carried by the same K -symbol codeword. Under semantic communications, SINR_N is directly used to evaluate the semantic similarity score $S_T(\text{SINR}_N)$ and the resulting semantic rate $R_{\text{sem},T}$ as detailed in Section III-C.

C. Semantic Communication Metrics

For a semantic link, the receiver evaluates the performance using semantic-specific variables rather than bit error rates. The three metrics below – semantic similarity, semantic rate, and semantic distortion – are evaluated identically for every link in the network (downlink near- and far-user links, uplink V2I, V2X, and eavesdropper links) by substituting the corresponding link SINR, so that a single consistent semantic performance model applies throughout the report.

1) *Semantic Similarity Score $S(\text{SINR})$* : The similarity between the original sentence's semantic features and the reconstructed features at the receiver is modeled by a sigmoidal function:

$$S(\text{SINR}) = \frac{1}{1 + \exp(-(\lambda_1 \text{SINR}_{\text{dB}} + \lambda_2))}, \quad (20)$$

where $\text{SINR}_{\text{dB}} = 10 \log_{10}(\text{SINR})$ and λ_1, λ_2 are scaling parameters capturing the linguistic complexity and compression capability of the deep semantic model. Intuitively, $S(\text{SINR}) \in (0, 1)$ saturates toward unity as the link SINR grows large, reflecting near-perfect meaning recovery under favorable channel conditions, and decays toward zero as SINR degrades, reflecting the collapse of semantic fidelity when the decoder can no longer reliably infer the transmitted concepts from the received symbols. The steepness of this transition is controlled by λ_1 , while λ_2 shifts the SINR operating point at which the decoder transitions from unreliable to reliable semantic recovery.

2) *Semantic Rate R_{sem}* : The semantic transmission rate, representing the number of semantic units transmitted per second, is defined as:

$$R_{\text{sem}} = B \cdot \left(\frac{M}{K} \right) \cdot S(\text{SINR}) \cdot \log_2(1 + \text{SINR}) \quad (\text{suts/sec}), \quad (21)$$

where B is the link bandwidth, M is the number of semantic concepts in a semantic unit, K is the number of transmitted channel symbols, and M/K represents the semantic compression ratio introduced at the beginning of Section III. This expression couples three distinct effects: the classical Shannon-type term $\log_2(1 + \text{SINR})$, which bounds the raw information-carrying capacity of the channel at the semantic-effective SINR defined in Section III-B; the semantic similarity factor $S(\text{SINR})$, which

discounts that capacity by the fraction of transmitted symbols that are actually decoded into correct meaning; and the explicit compression ratio M/K , which converts the resulting channel-symbol rate into a rate of *semantic units* per second. Note that M/K therefore enters R_{sem} twice, in two distinct roles: implicitly, through the K/M dilution already embedded in SINR (which lowers both $\log_2(1 + \text{SINR})$ and $S(\text{SINR})$ as compression increases), and explicitly, as the units-conversion multiplier M/K in front. This combination produces the trade-off anticipated in Section III: increasing M/K raises R_{sem} through the explicit multiplier, but simultaneously lowers the semantic-effective SINR and hence $S(\text{SINR})$ and $\log_2(1 + \text{SINR})$ through the implicit K/M dilution; the semantic rate is therefore maximized at an intermediate compression ratio rather than growing without bound as compression is increased, consistent with the qualitative discussion given earlier in this section.

3) *Semantic Distortion D_{sem}* : The average loss of semantic meaning due to channel impairments is quantified as:

$$D_{\text{sem}} = 1 - S(\text{SINR}). \quad (22)$$

Since $S(\text{SINR})$ already reflects the joint effect of channel quality and compression aggressiveness through the semantic decoder's operating characteristic, D_{sem} directly captures the residual meaning that is lost at the receiver, whether due to weak SINR, high jamming interference, or an overly aggressive semantic compression ratio M/K chosen at the encoder.

D. Uplink Semantic Communication

The Target T also acts as an active terrestrial transmitter, encoding its uplink messages through the same semantic encoder pipeline described at the start of Section III before transmission.

- **Uplink V2I (Target to BS via RIS R)**: The Target transmits its semantic signal with power P_T . The cascaded channel gain from the Target to the BS B via the RIS R is $g_{T,R,B}$, and $P_{J,B} = P_J g_{J,B}$ is the received jamming interference at the BS B from Jammer J . The conventional link-budget ratio is $\Gamma_{\text{uplink}} = \frac{P_T g_{T,R,B}}{P_{J,B} + \sigma^2}$, so that the semantic-effective SINR at the BS B is:

$$\text{SINR}_{\text{uplink}} = \left(\frac{K}{M} \right) \cdot \frac{P_T g_{T,R,B}}{P_{J,B} + \sigma^2}. \quad (23)$$

- **V2X Direct Semantic Link (Target to Mobile User U)**: The Target communicates directly with the Mobile User U via a ground link with channel gain $g_{T,U}$. The conventional link-budget ratio is $\Gamma_{\text{V2X}} = \frac{P_T g_{T,U}}{P_{J,U} + \sigma^2}$, so that the semantic-effective SINR at the Mobile User U is:

$$\text{SINR}_{\text{V2X}} = \left(\frac{K}{M} \right) \cdot \frac{P_T g_{T,U}}{P_{J,U} + \sigma^2}. \quad (24)$$

For both links, the same K/M compression-induced dilution applies as in the downlink, since the Target's uplink and V2X messages are generated by the same semantic encoder pipeline and carried over the same K -symbol semantic codeword length. The semantic similarity, rate, and distortion are then evaluated using the corresponding formulas in Section III-C, with the same compression ratio M/K and decoder parameters λ_1, λ_2 applying consistently across the uplink, V2X, downlink, and eavesdropper channels.

E. Monostatic Radar Target Sensing at UAV Jammer

In addition to transmitting interference, the UAV Jammer J performs direct monostatic radar sensing to detect and track both the Target T and the eavesdroppers E_i separately. To ensure that these sensing processes do not intermix or cause mutual interference, the Jammer J transmits orthogonal sensing waveforms (or uses spatially separated beamforming). This allows the Jammer to sense the Target T and the eavesdroppers E_i separately, ensuring that they do not intermix.

The received sensing SNR at the Jammer J for the Target T is:

$$\text{SNR}_{\text{radar},T} = \frac{P_{J,\text{sense}} \cdot g_{J,T,J}}{\sigma^2}, \quad (25)$$

where $P_{J,\text{sense}}$ is the dedicated sensing transmit power allocated by the Jammer J for Target T , and $g_{J,T,J} = g_{J,T}^2$ is the round-trip monostatic channel gain.

Similarly, the received sensing SNR at the Jammer J for eavesdropper E_i is:

$$\text{SNR}_{\text{radar},E_i} = \frac{P_{J,\text{sense}} \cdot g_{J,E_i,J}}{\sigma^2}, \quad (26)$$

where $g_{J,E_i,J} = g_{J,E_i}^2$ is the round-trip monostatic channel gain for eavesdropper E_i . Because of the orthogonal signals, they do not intermix.

The corresponding semantic sensing accuracies for Target T and eavesdropper E_i are modeled as:

$$A_{\text{sensing},T} = \frac{1}{1 + \exp(-\kappa(\text{SNR}_{\text{radar},T,\text{dB}} - \tau))}, \quad (27)$$

$$A_{\text{sensing}, E_i} = \frac{1}{1 + \exp(-\kappa(\text{SNR}_{\text{radar}, E_i, \text{dB}} - \tau))}, \quad (28)$$

where $\text{SNR}_{\text{radar}, k, \text{dB}} = 10 \log_{10}(\text{SNR}_{\text{radar}, k})$ for $k \in \{T, E_i\}$, $\kappa = 0.4$ represents the classification slope, and $\tau = 2.0$ dB represents the minimum SNR threshold for target detection.

F. Eavesdropper Wiretap Model

The non-colluding eavesdroppers E_i ($i \in \{1, 2, 3\}$) attempt to wiretap the downlink transmission. Eavesdropper E_i wiretaps the superposed signal, targeting the far user signal s_F (which has a higher power allocation factor). The conventional link-budget ratio at eavesdropper E_i is $\Gamma_{E_i} = \frac{a_F P_B g_{E_i}}{a_N P_B g_{E_i} + P_{J, E_i} + \sigma^2}$, where $g_{E_i} = g_{\text{cascaded}, E_i} + g_{\text{direct}, E_i}$ is the total effective channel gain from the BS B to eavesdropper E_i , and $P_{J, E_i} = P_J g_{J, E_i}$ is the jamming power received at eavesdropper E_i from Jammer J . Since the eavesdropper intercepts the same K -symbol semantic codeword as the legitimate users, the identical compression-induced scaling applies, giving the semantic-effective wiretap SINR:

$$\text{SINR}_{E_i} = \left(\frac{K}{M} \right) \cdot \frac{a_F P_B g_{E_i}}{a_N P_B g_{E_i} + P_{J, E_i} + \sigma^2}. \quad (29)$$

The corresponding semantic similarity score $S_{E_i}(\text{SINR}_{E_i})$ and semantic wiretap rate R_{sem, E_i} are computed at each eavesdropper. It is worth emphasizing that semantic communication provides a security benefit that is structurally distinct from, and complementary to, conventional SINR-based secrecy, and that this benefit now compounds with the K/M dilution built into SINR_{E_i} itself: even where an eavesdropper E_i achieves a non-negligible semantic-effective SINR_{E_i} and successfully captures raw semantic symbols, meaningfully reconstructing the transmitted message additionally requires access to a semantic decoder matched to the legitimate transmitter's encoder (i.e., the same learned mapping from symbols to concepts). Absent this shared semantic model, the intercepted symbols carry little exploitable meaning, so that the semantic similarity score $S_{E_i}(\text{SINR}_{E_i})$ – and hence the wiretap rate R_{sem, E_i} – systematically underestimates the true semantic leakage an eavesdropper without the correct decoder would experience in practice. This effect is in addition to, not a substitute for, the jamming- and compression-induced degradation already captured in SINR_{E_i} . The system security is characterized by ensuring that the legitimate semantic rate is strictly greater than the wiretap rate:

$$R_{\text{sem}, U} > R_{\text{sem}, E_i}, \quad \forall i. \quad (30)$$

IV. CONCLUSION AND FUTURE DIRECTIONS

In this weekly report, we presented the complete system architecture and mathematical foundations of the Week 9 semantic-aware ISAC network. The physical propagation links, PD-NOMA coding with semantic-aware SIC operating on reconstructed semantic representations rather than raw bit streams, semantic metrics (similarity, rate, distortion) governed by a consistent compression ratio M/K across all links, uplink V2I/V2X links, jammer radar sensing, and multi-eavesdropper topologies have been successfully integrated into a cohesive, object-oriented Python implementation.